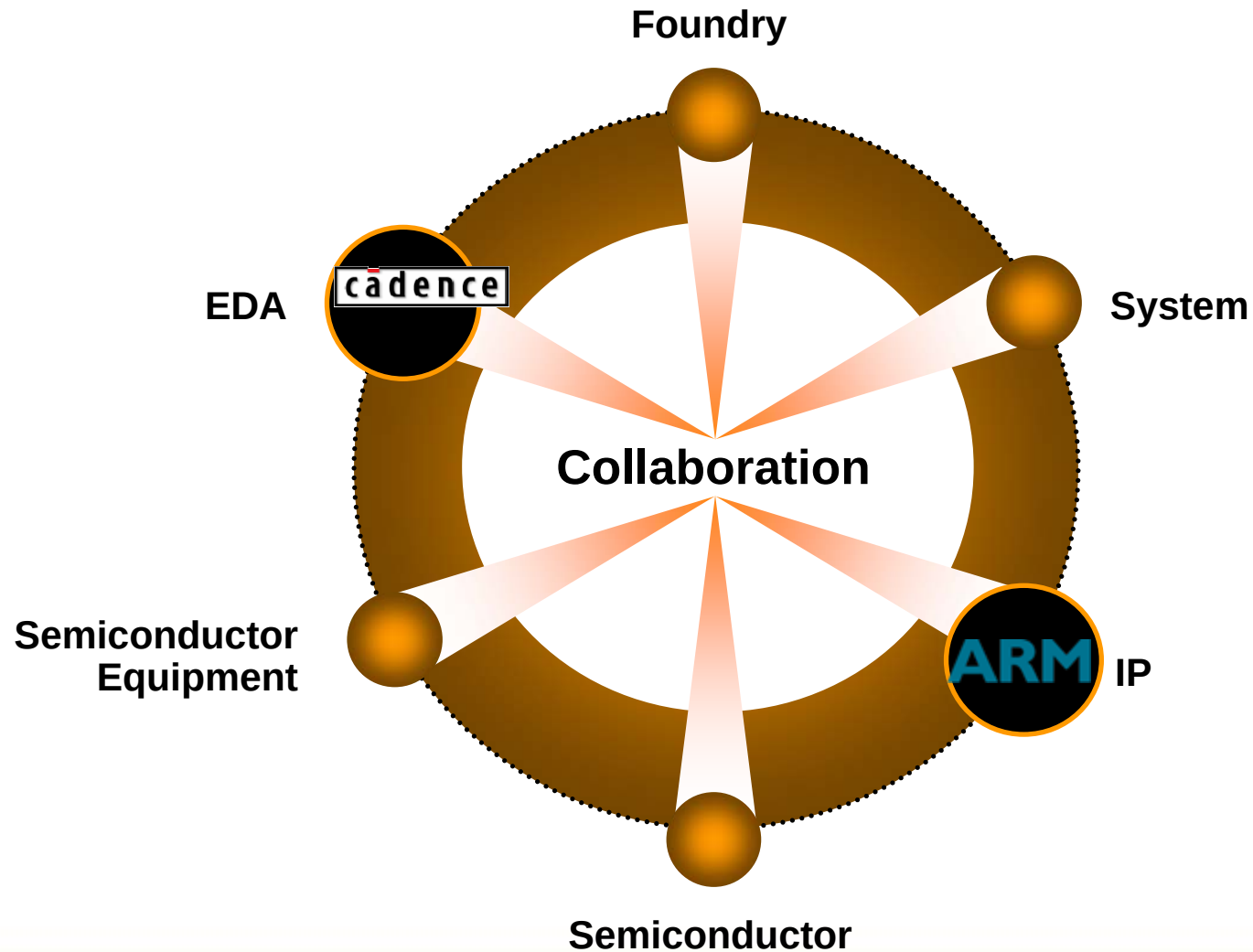




SILICON DESIGN CHAIN
APPLIED MATERIALS | ARM | ARTISAN | CADENCE | TSMC

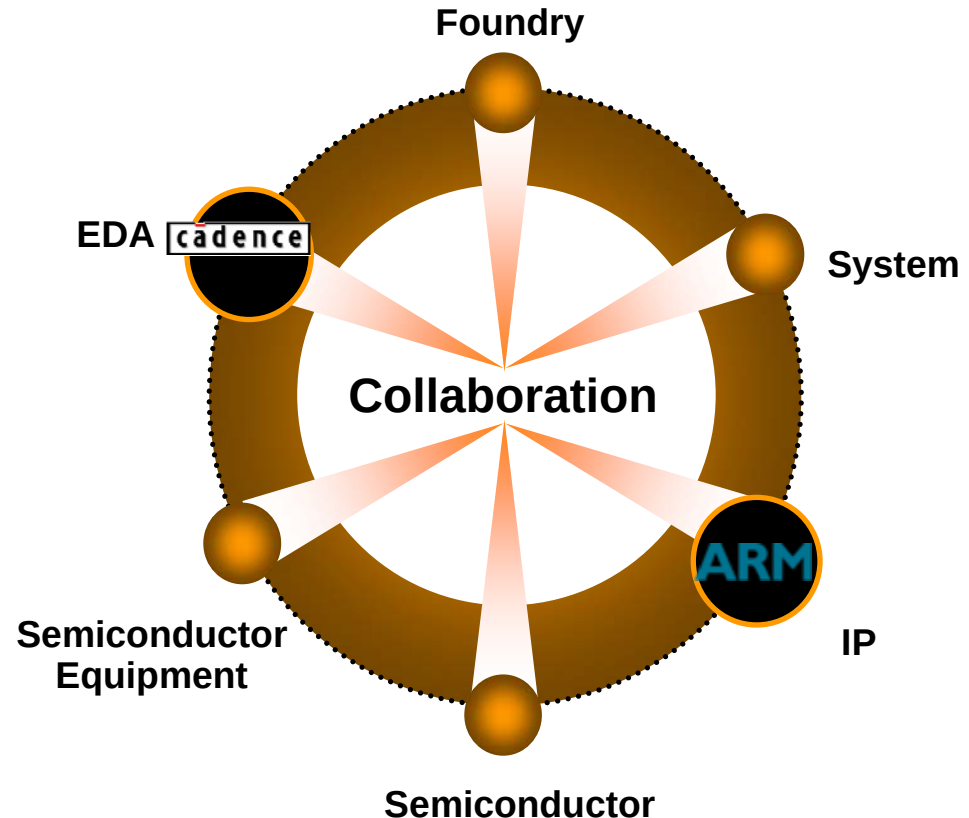
ARM and Cadence Collaboration in the Silicon Design Chain

Collaboration is No Longer 1:1



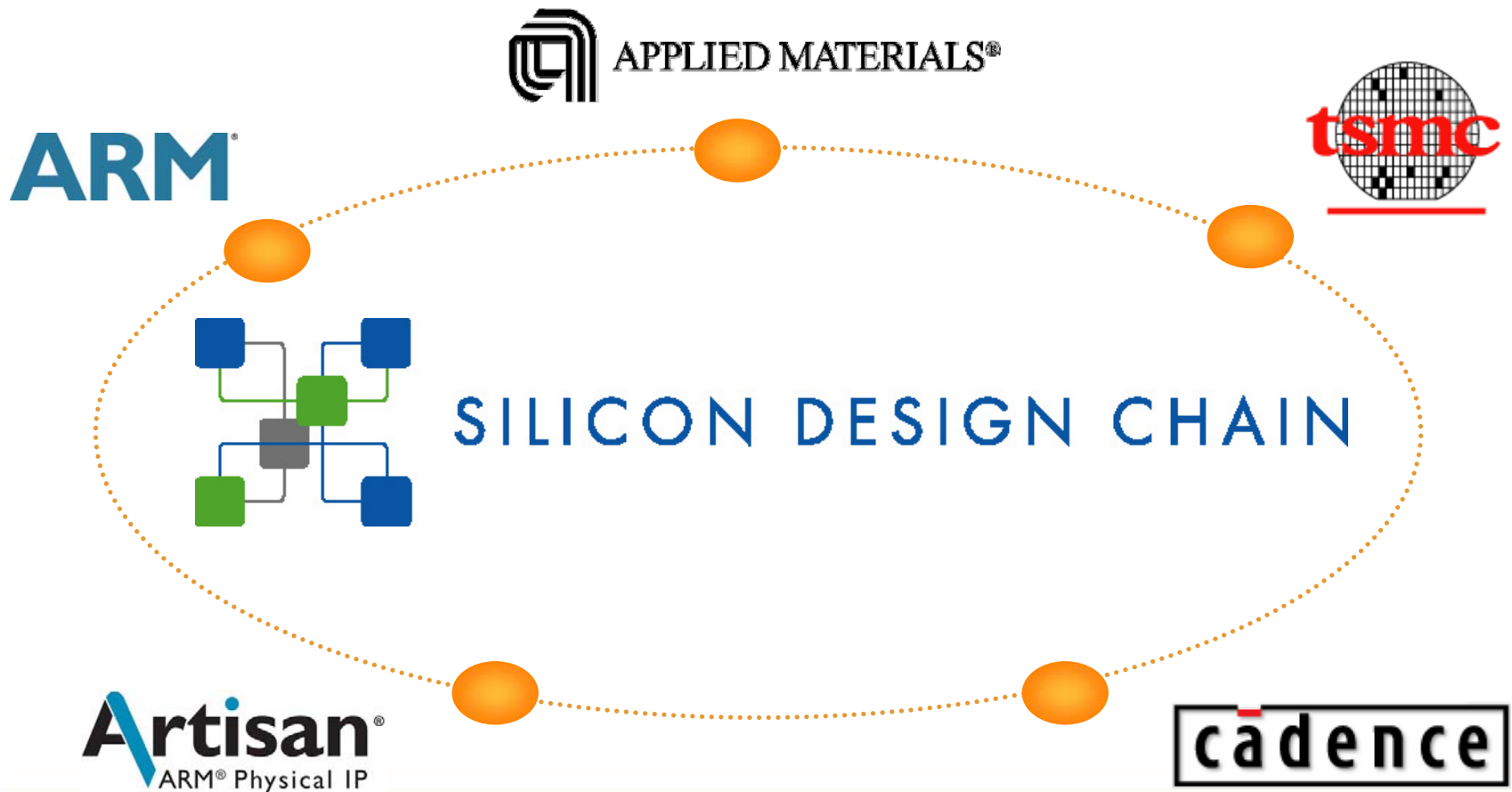
ARM-Cadence Collaboration

- Focused programs
 - SDC low-power test chip
 - ARM-CDN Encounter Reference Methodology
 - Cadence® Optimization Methodology Kit for ARM® Processors
- Delivering customer benefits
 - Improved time to market
 - Lower design risk
 - Flexible, differentiated design solutions



Silicon Design Chain Alliance

Industry Leaders Team to Address Top Cross-Industry Issues

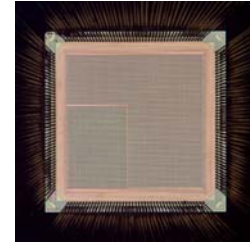


SILICON DESIGN CHAIN

SDC Proof Point Project

90nm Low-Power Test Chip

- Vehicle: ARM processor-based test chip
 - ARM1136JF-S™
 - TSMC 90nmG process
 - ARM Artisan® SAGE-X3 library
 - ARM Power-Management Kit
- Goal: Validate RTL-GDSII power management flow
- Silicon validated results
 - First silicon success – working chip/board
 - RealView® family-core module
 - **40% power reduction verified**



SDC Collaboration Visibility

- Industry awareness of the project success and value:
 - 3+ conference papers, 26 articles, IEEE CICC paper(2005)

"Initiative brings low power to mainstream design"; EE Times; March 21, 2005 - PRINT ALSO.
<http://www.eet.com/news/design/showArticle.jhtml?articleID=159902153>

"Four industry players team up to address power issue"; EE Times Asia; March 24, 2005;
http://www.eetasia.com/ART_8800362403_fa801c99_no.HTM

EE Times - SDC low-power initiative / Joint contributed article by George Kuo and ARM: "Tackling test challenges for low-power design"; November 7, 2005;
<http://www.eetimes.com/news/design/showArticle.jhtml?articleID=173403100&pgno=1>

EE Times - How to use power gating techniques:
<http://www.eetimes.com/news/latest/showArticle.jhtml?articleID=174900285>

EE Times; "Panelists optimistic on lower power design"; November 4, 2005;
<http://www.eetimes.com/news/design/showArticle.jhtml?articleID=173402884>

EE Times; "Panel studies power in SoCs"; November 4, 2005;
<http://www.eetimes.com/news/design/showArticle.jhtml;jsessionid=NTU3QVXUHW1O4QSNDBCCKHSCJUM EKJVN?articleID=173500011>

White Paper:

"Low-power flow enables multi-supply voltage ICs" by Rob Aitken (ARM Physical IP), George Kuo (Cadence) and Ed Wan (TSMC); EE Times; March 21, 2005
<http://www.eetimes.com/news/design/showArticle.jhtml?articleID=159902216>



SDC Test Chip Results

- ARM1136JF-S core design demonstrated successful balance of performance with power management
 - Taped out at 355 MHz
 - With much lower power

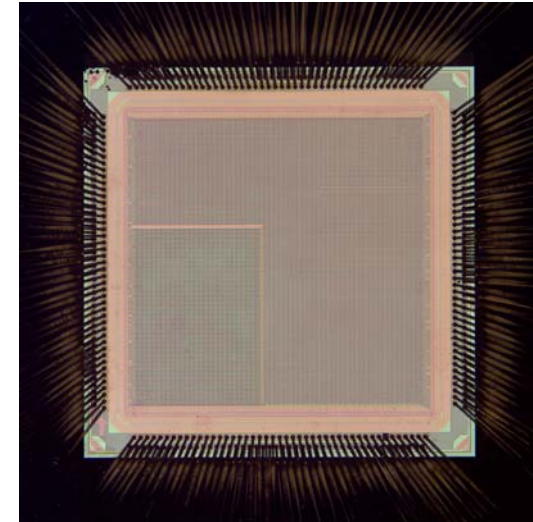
	Top (1.0V) domain	NoRam(0.8V) domain	Subtotal
Dynamic power savings	12.0%	50.3%	37.9%
Leakage power savings	69.3%	33.6%	46.7%
Chip total savings			40.3%

**Generalized and effective power management strategy
delivered > 40% savings !**



Silicon Validated Power Savings

Domain	Simulated 90nm Baseline (mW/MHz)	Simulated 90nm Low Power (mW/MHz)	Measured 90nm Low Power (mW/MHz)	130nm Reference ¹ (mW/MHz)
Core (0.8V)	0.28	0.14	0.10	0.6
Other (1.0V)	0.36	0.32	0.21	



First silicon success, with > 85% yield

Note:

1. Published power number for ARM1136JF-S core, based on 130nm technology and under typical condition:
ARM public website <http://www.arm.com/products/CPUs/ARM1136JF-S.html>



SILICON DESIGN CHAIN

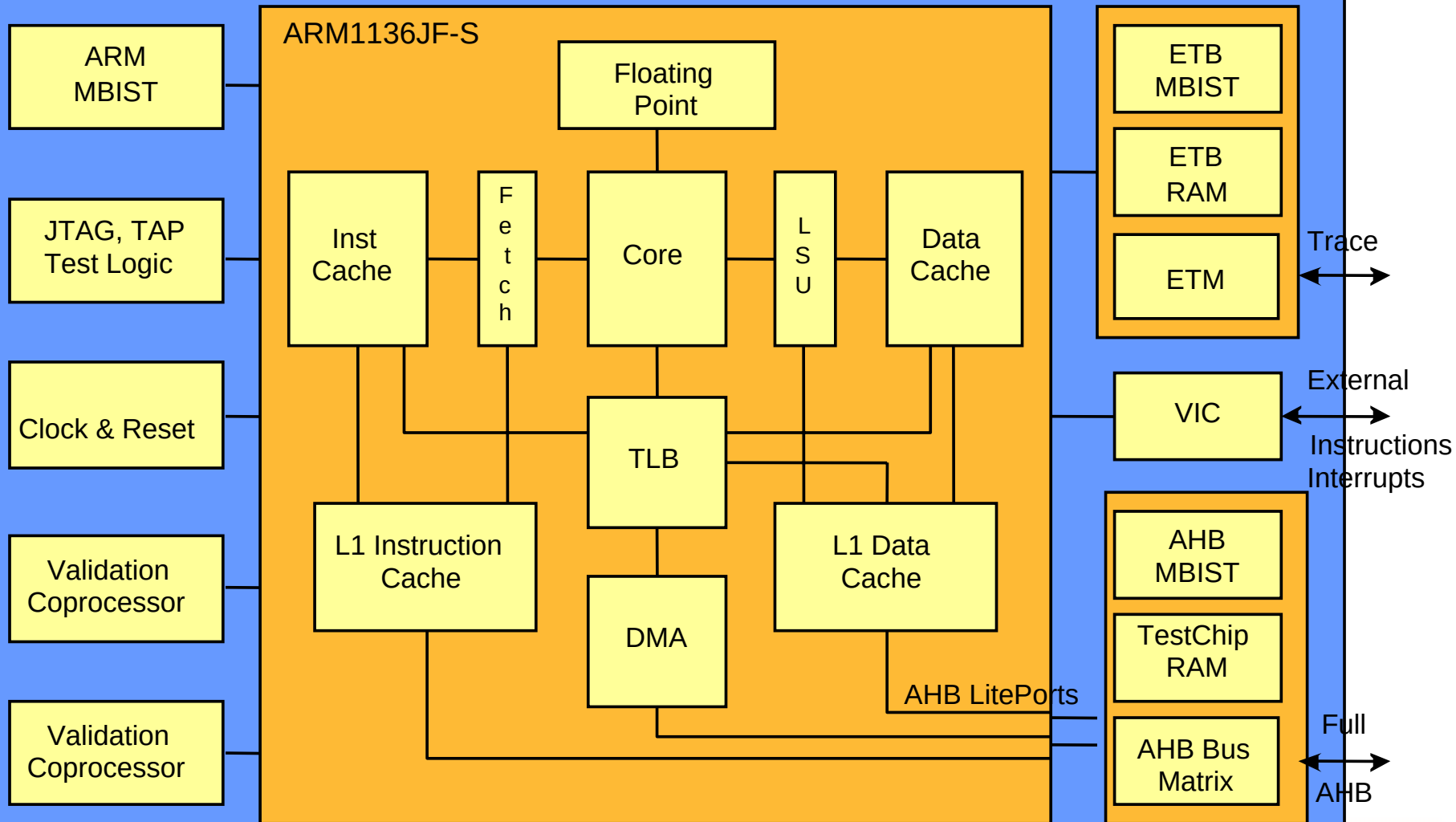
Technology Demonstration

1. Leakage power optimization
 - Multi- V_{th} capability
2. Dynamic power optimization
 - Multi- V_{dd} capability (MSV)
 - Automating advanced clock gating
3. ECSM supporting multi- V_{dd} design
 - Signal Storm, CeltIC → Extremely fast characterization (10X SPDM), Within 1% of SPICE, Supports unlimited voltage gradations
4. Power analysis and correlation
5. Cohesive low-power solution throughout the design flow
6. Silicon validated power saving

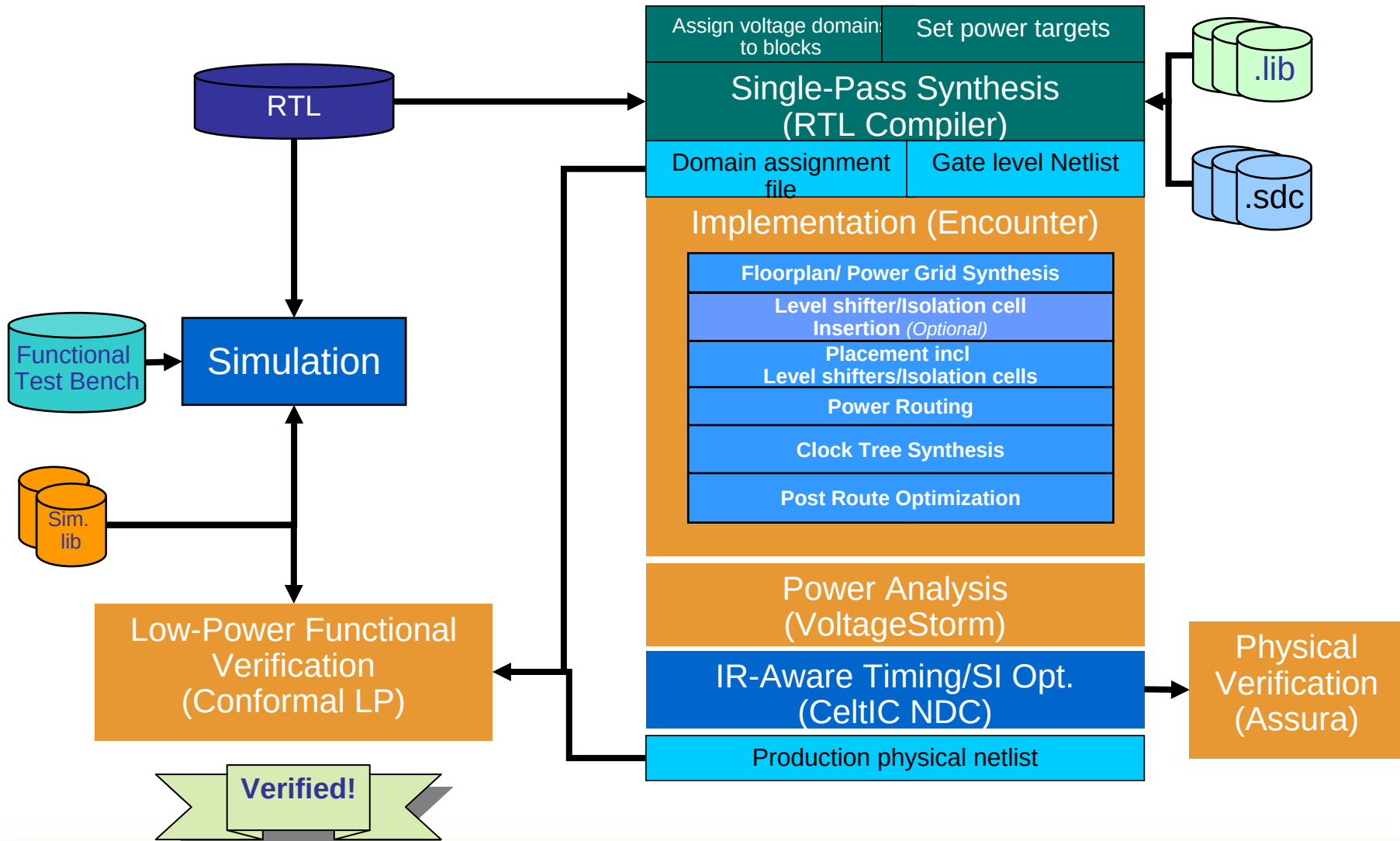


ARM1136 Test Chip Architecture

ARM1136_TC

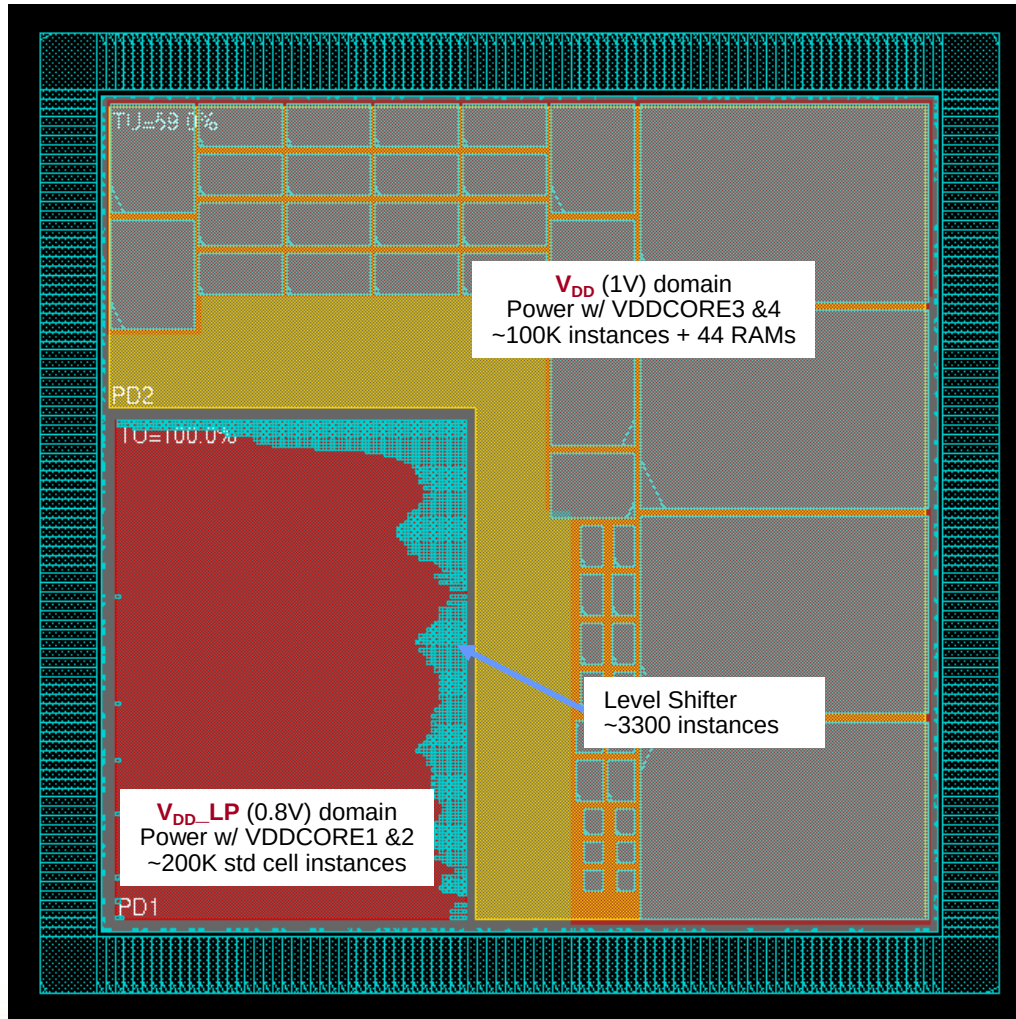


Low-Power Design Flow



Low-Power Implementation

Physical implementation results



Physical information

- Die size 4mm x 4mm
- # of pads = 360
- Cell count ~300K
- Utilization ~80%
- Memories ~60% of area

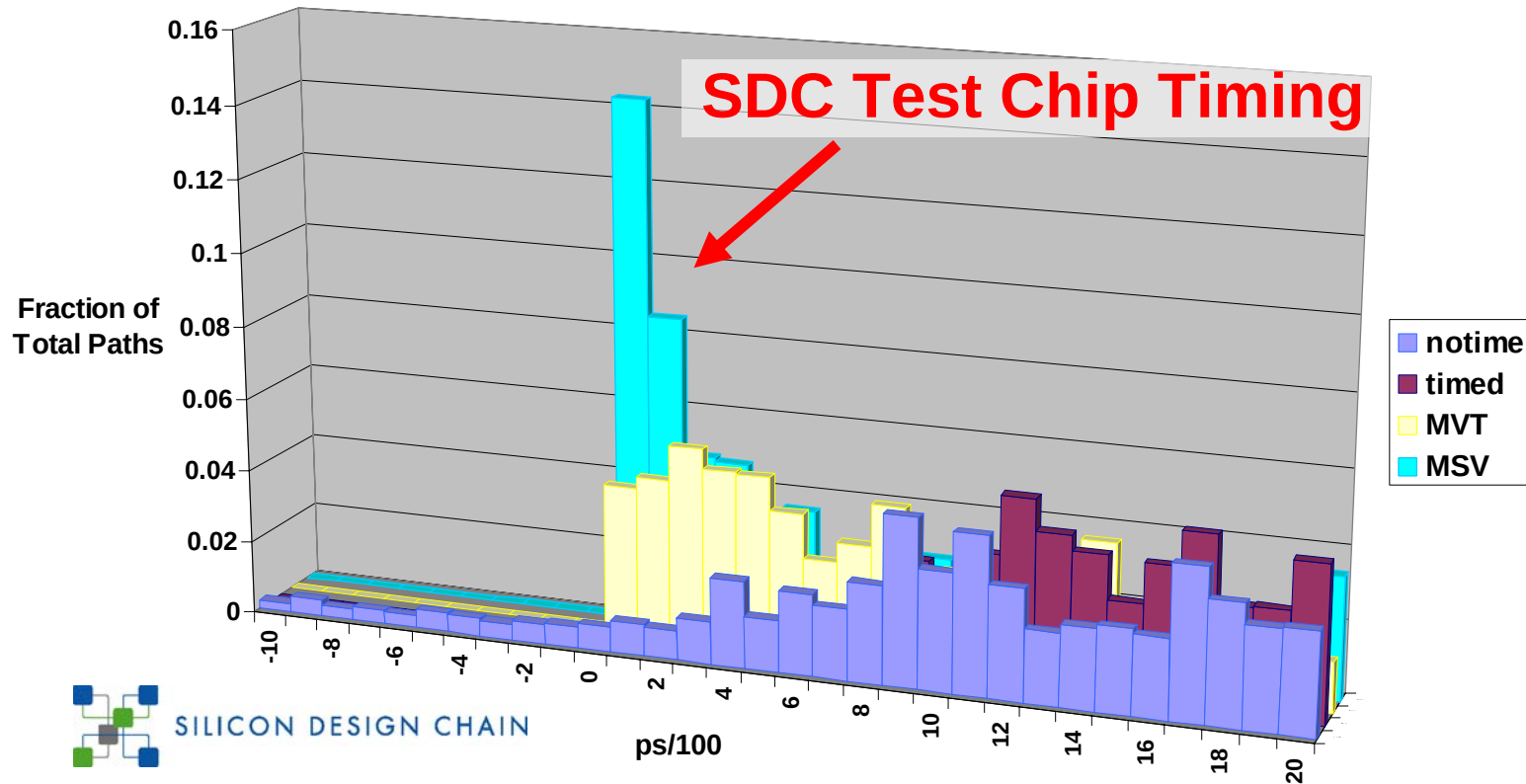
Low-power implementation

- Two power domain: 1.0V and 0.8V
- Level shifters for signal conditioning
- Multi-V_{th} cell usage for leakage power reduction
- Extensive clock gating throughout: ~85% registers were gated



Low-Power Design Effects DFT

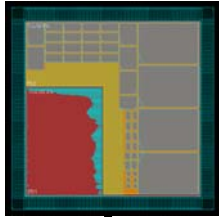
Test chip Timing Path Slack Distribution -100ps -> +200ps



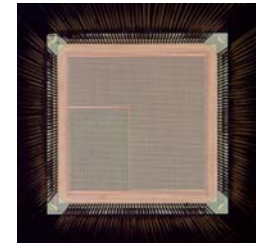
Low-Power Design Techniques Increase Timing Criticality



Test and Characterization



**TSMC 90G
Process**



Design planning/implementation
Front End Design For Test
RC/Encounter Test

True-Time
Delay Test

STIL (IEEE 1450)
Vectors

DFT Tester

Logical/physical x-reference

Fault models

ATPG

Test vectors
▪ stuck-at
▪ transition

Failure patterns
▪ passed tests
▪ failed tests

Encounter
Diagnostics

Callout
report

- Scan
- BIST*
- JTAG

Volume Diagnostics
Precision Diagnostics

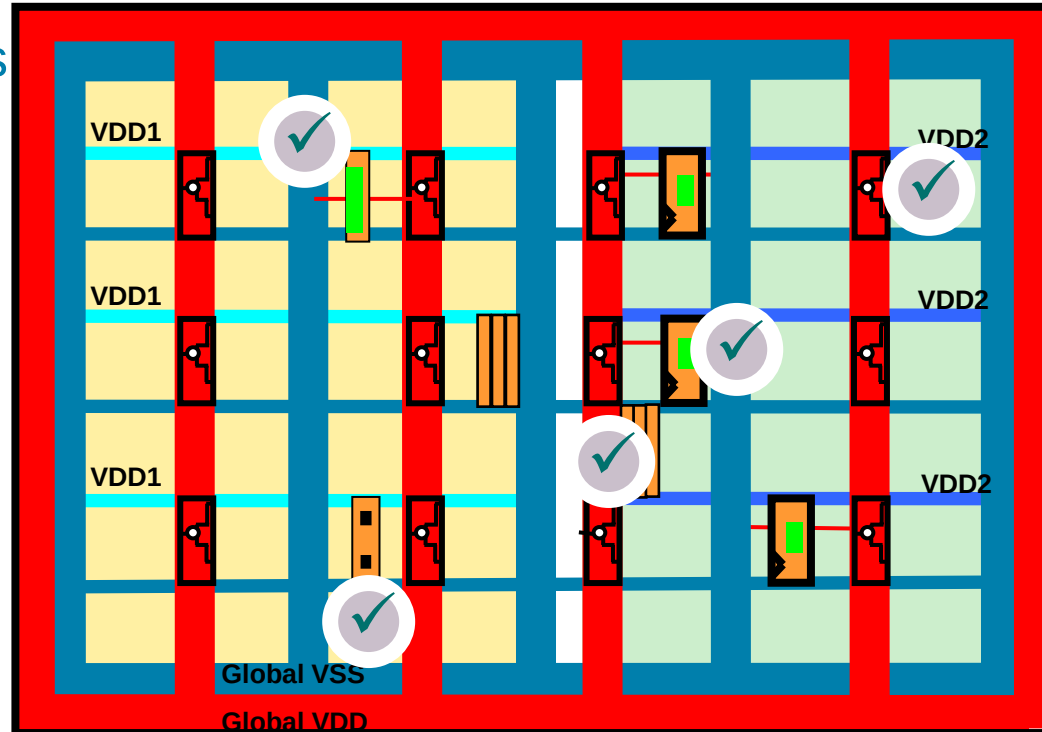


SILICON DESIGN CHAIN

ARM Power-Management Kit

- Power gates*
 - Power control of voltage islands via switchable voltage rails using header (shown) or footer cells
- Level shifter and isolation cells
 - Up and down shifting with optional enable signal
- Retention flip-flops*
 - Maintain FF state after power down for leakage reduction
- Back-bias support*
 - Reduce leakage current via well-biasing with special fill_tie cells
- Always on buffer*
 - Buffering of signals in powered-down areas

*Note: requires tapless library



Note: Picture shows a conceptual implementation



ARM Intelligent Energy Management (IEM)

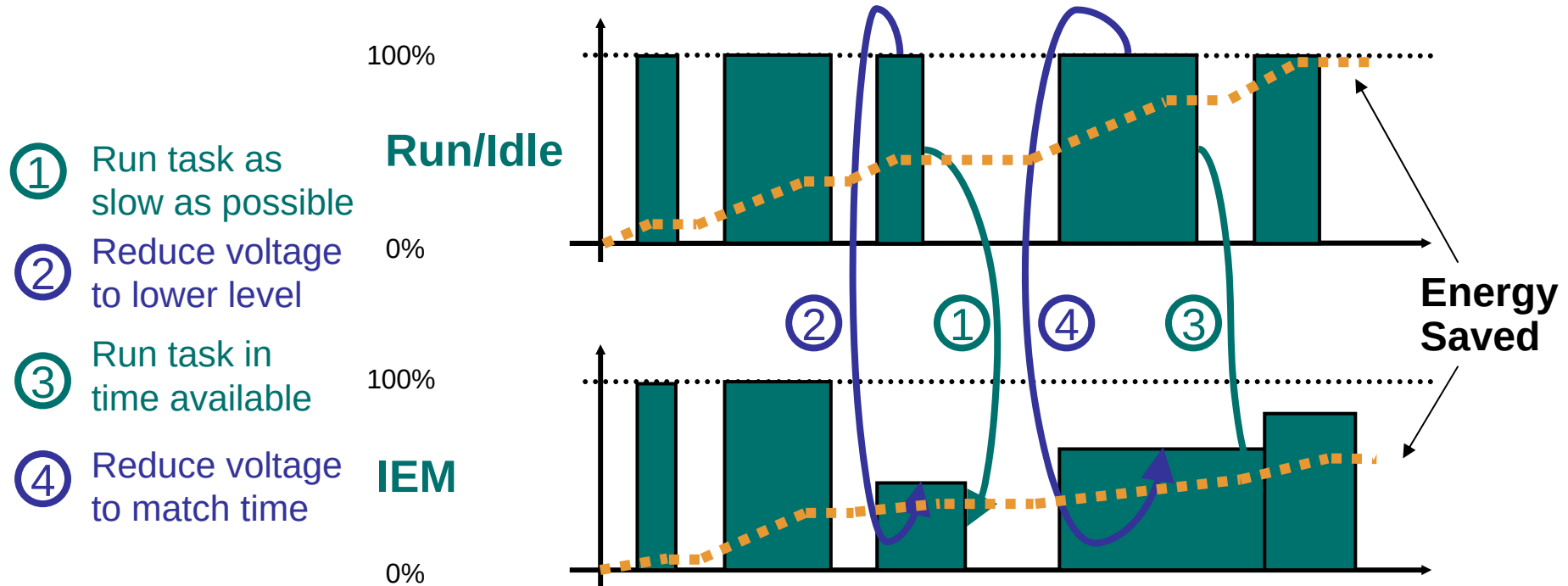
- IEM™ solution
 - Software, core, on-chip infrastructure, libraries, design flow, partnerships
 - Enables *OS-Based*, dynamic control of SoC power by self-modulation of V_{dd} (supply voltage) and Frequency
 - Uses off-chip regulator from NSC for multi-voltage operation
 - Switches between multiple PLLs for clock control
- IEM reduces power and energy consumption
 - Many applications do not require all of the processors performance
 - IEM uses Dynamic Voltage and Frequency Scaling (DVS) techniques
 - **Saves up to 60% on the energy used in an ARM powered SoC design**
 - **Result in an increase of product's battery life by 15-20%***
- IEM has no discernable impact upon user experience

* Estimate from ARM's lead OEM partner

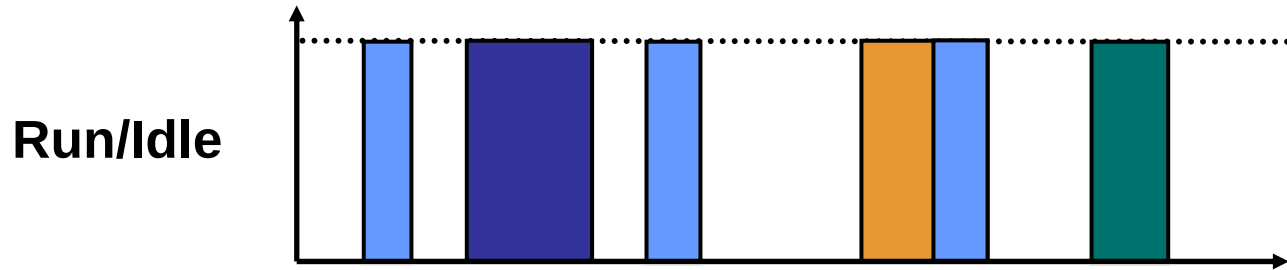


How Does IEM Reduce Power and Energy?

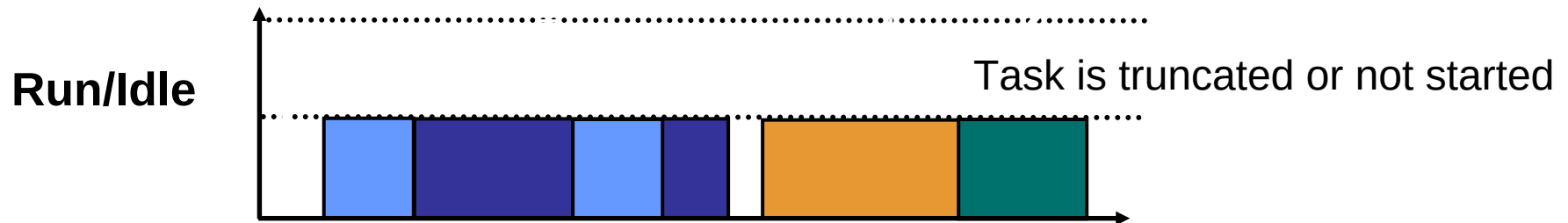
- Batteries contain finite energy
- Running fast and idling is wasteful:
 - Only need to run just fast enough to meet the application software deadlines and maintain quality



When to Change Frequency/Voltage?



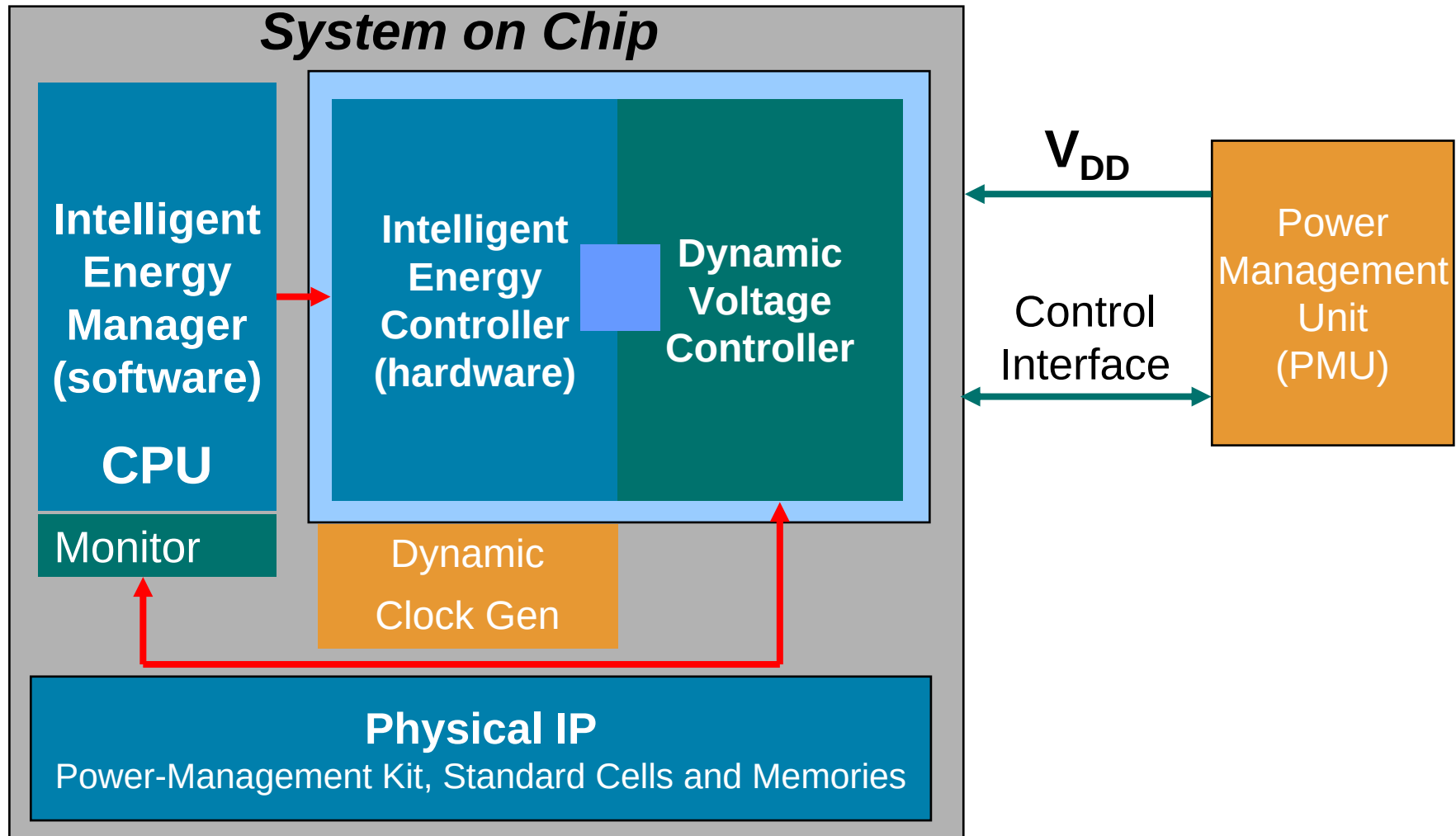
- How to detect frequency/voltage is too high
 - Excessive time spent in idle task
 - Long gaps before next tasks



- How to detect insufficient frequency/voltage
 - Idle task runs infrequently
 - Tasks are often being pre-empted
 - Frames are in danger of being dropped (audio, video, games)

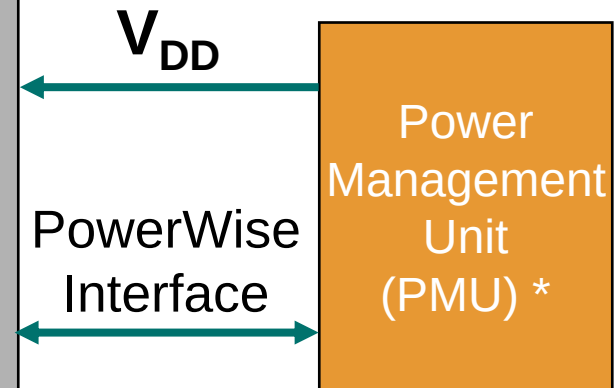
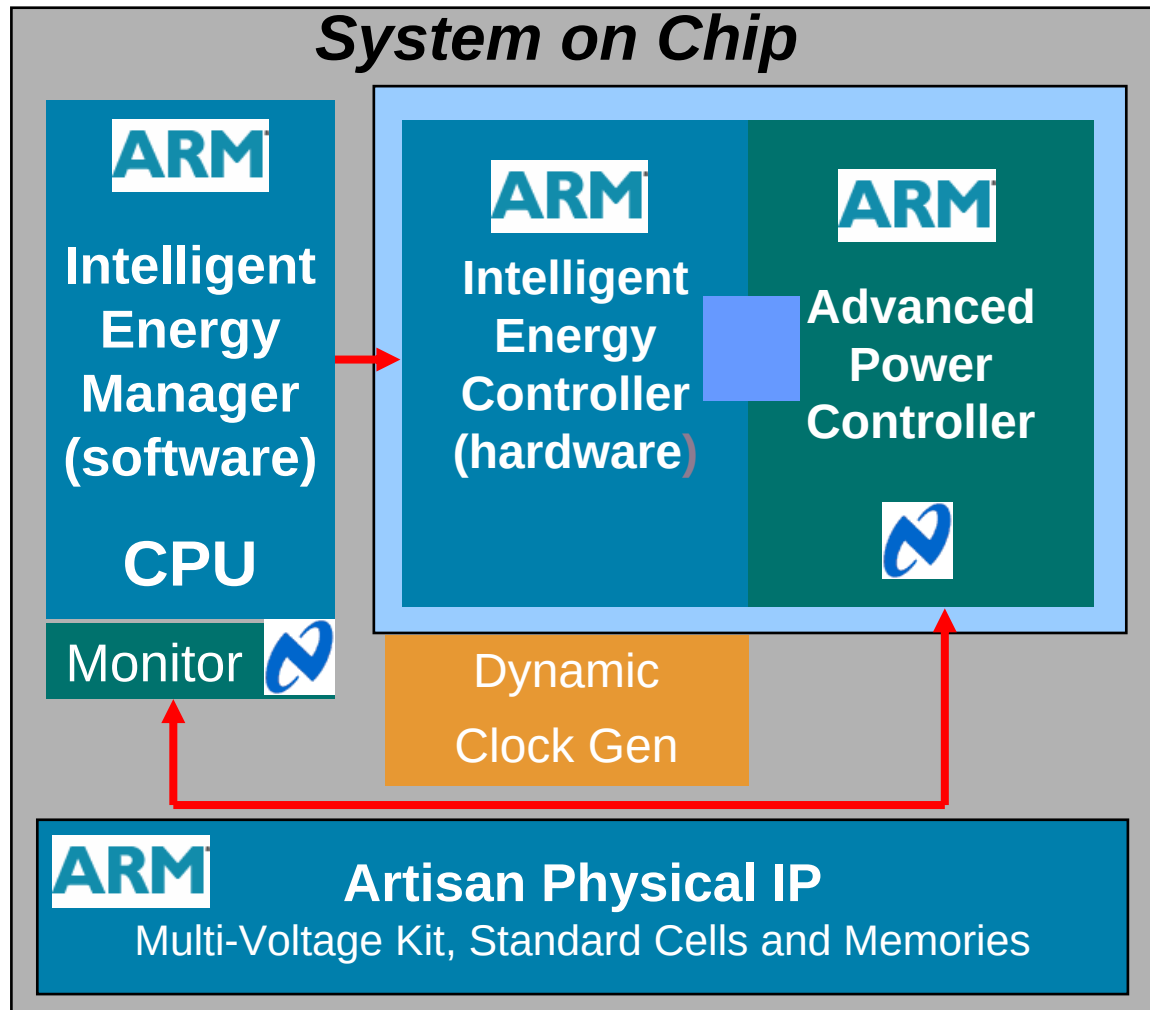


IEM Overview

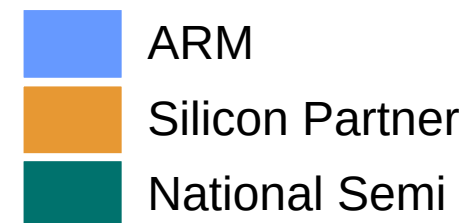


IEM Overview

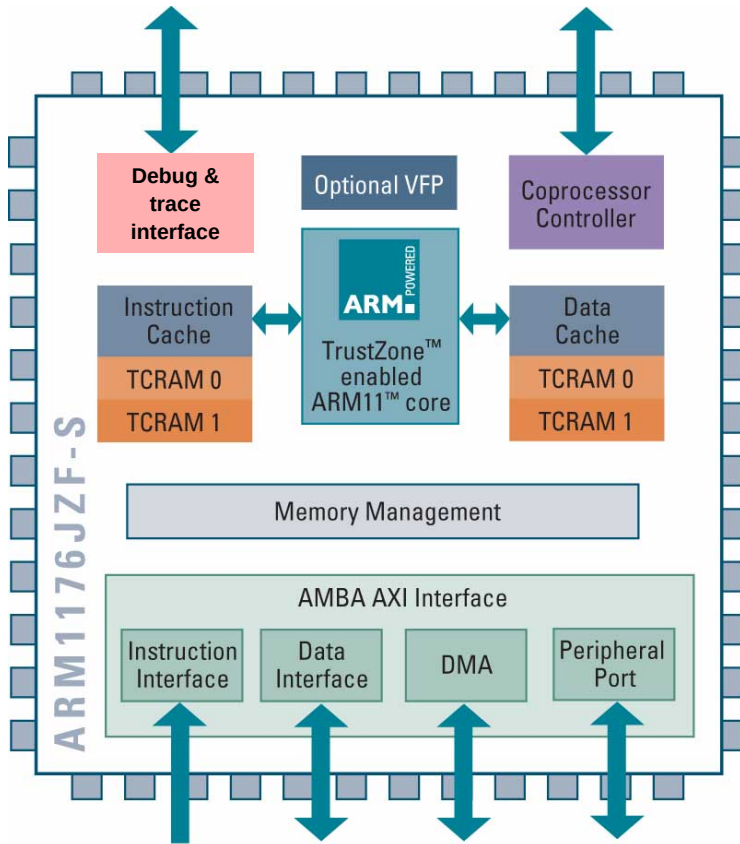
- APC includes Hardware Performance Monitor technology, which enables closed-loop DVS



* Slave Power Controller Licence available from National Semi



ARM1176JZ(F)-S Solution



**333 MHz on TSMC CL013G,
550 MHz on TSMC CL013HS+
500 MHz on TSMC CLN90G**

- **550MHz ARM11™ processor**
 - Fast, 8-stage pipeline
 - Dynamic branch prediction and return stack
 - Jazelle™ media extensions
 - Enhanced real-time performance
 - Optional vector floating point
- **Low power with IEM**
 - **Supports multiple voltage domains (Dormant Mode) and voltage scaling**
- **Core of secure system with ARM TrustZone™**
 - Architecture extensions for CPU and system security
- **High-performance AXI Memory system**
 - Configurable I & D caches and TCMs with dedicated 64-bit DMA interface
- **Reduced software development effort**
 - On-chip CoreSight™ debug/trace infrastructure
 - Models, tools, boards
- **Best in class developer support**
 - PrimeXsys® platform – best possible starting point
 - Reference methodology with leading EDA vendors



IP Deployment Challenge - Hard-IP or Soft-IP?

Hard-IP is predictable

- Known characteristics
- Generic implementation
- Fully verified
- Enables reuse without re-verification

Soft-IP is flexible

- Process portable
- Configurable
- Application-specific implementation
- Supports user-specific IP diagnosis

But... predictability can be restrictive

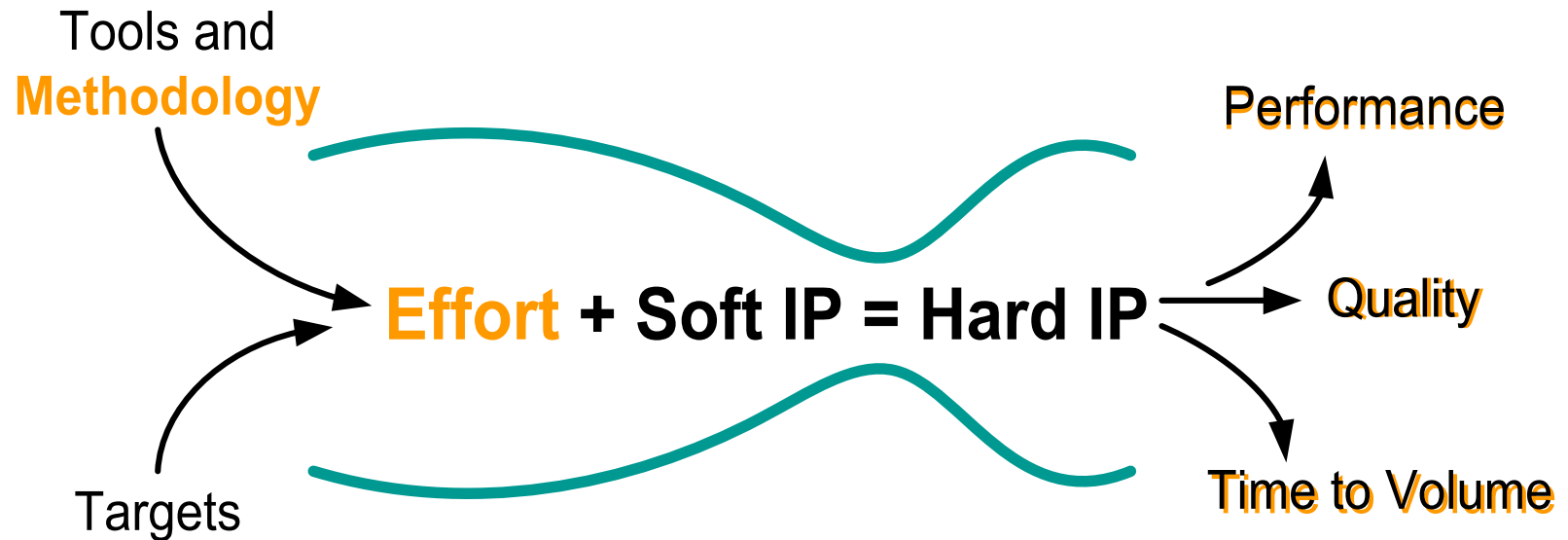
- Fixed technology
- Fixed implementation
- No application-specific consideration

But... flexibility can be a liability

- Requires re-verification
- Unknown characteristics
- Larger overhead for reuse



The Best of Both Worlds



***ARM Implementation Reference Methodology:
A proven path for the deployment of ARM IP
with best-in-class EDA tools***

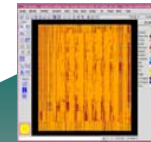


ARM-Cadence Encounter Reference Methodology

- Scripted RTL to GDSII flow delivers
 - Optimised PPA
 - Reduced turnaround time
 - Risk/cost reduction
- Key elements
 - Core hardening
 - Generation of abstract models
- Available from ARM and Cadence
 - Jointly developed flows
 - Supports ARM7EJ-S™, ARM7TDMI-S™, ARM926EJ-S™, ARM946E-S™, ARM966E-S™, ARM968E-S™, ARM1026EJ-S™, ARM1136J(F)-S
 - ARM1156T2(F)-S™, ARM1176JZ(F)-S™ in Q106
 - ARM11™-MPCORE™, Cortex-M3™ in H106

ARM endorsed flow for ARM soft IP implementation using Encounter

Silicon Ready Implementation



**ARM-Cadence
Encounter
Reference
Methodology**

**Encounter
Platform**

cadence

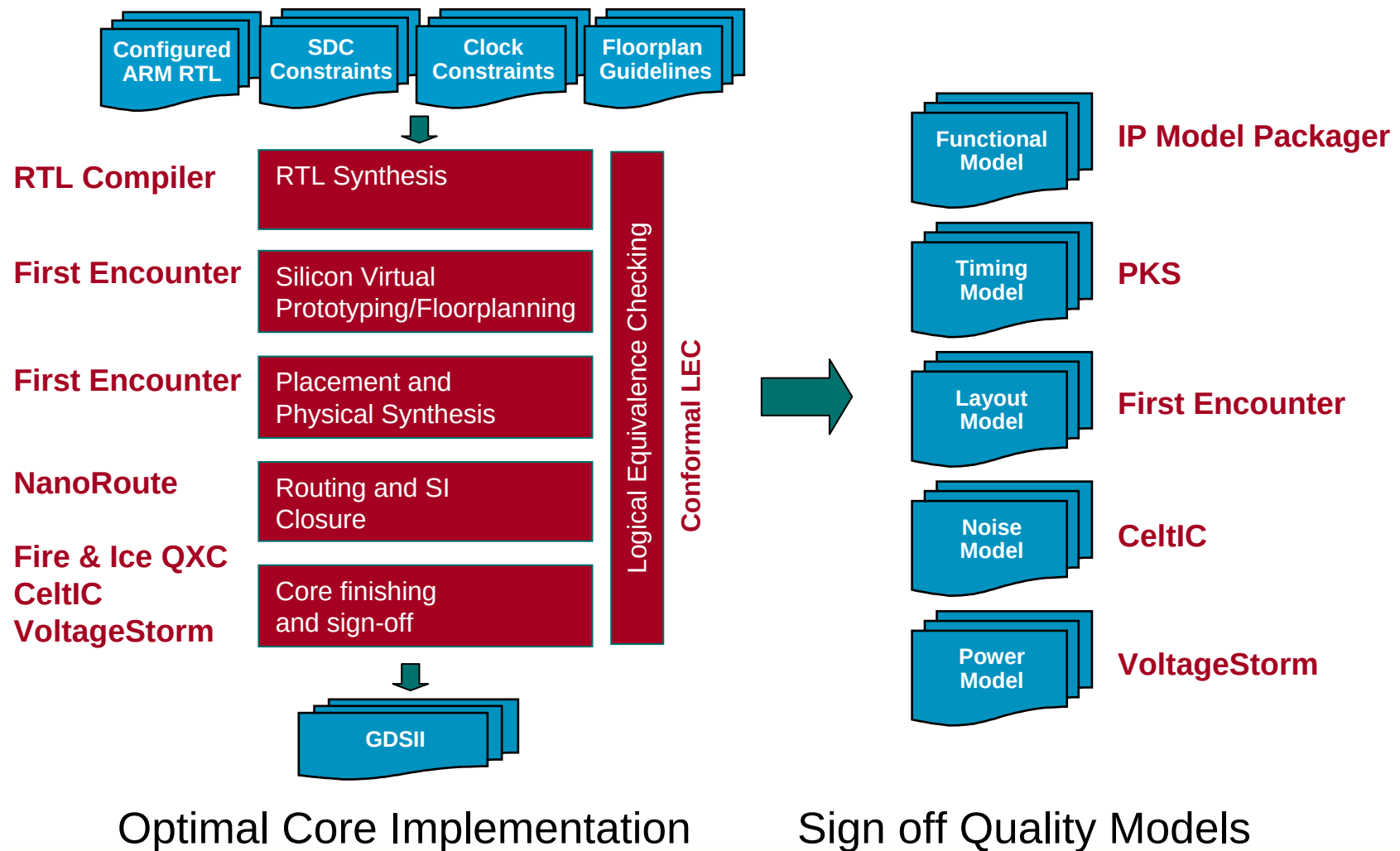
ARM Soft IP

ARM



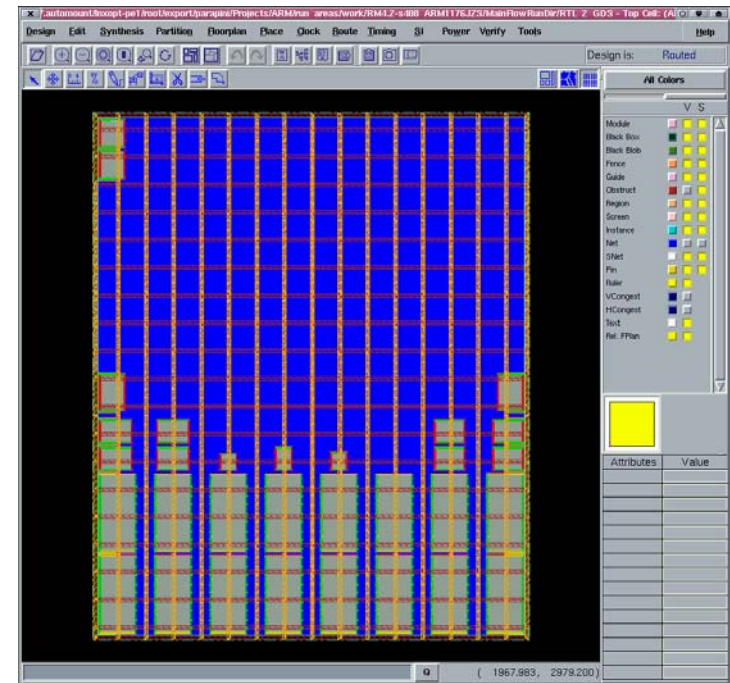
SILICON DESIGN CHAIN

Reference Methodology Flow



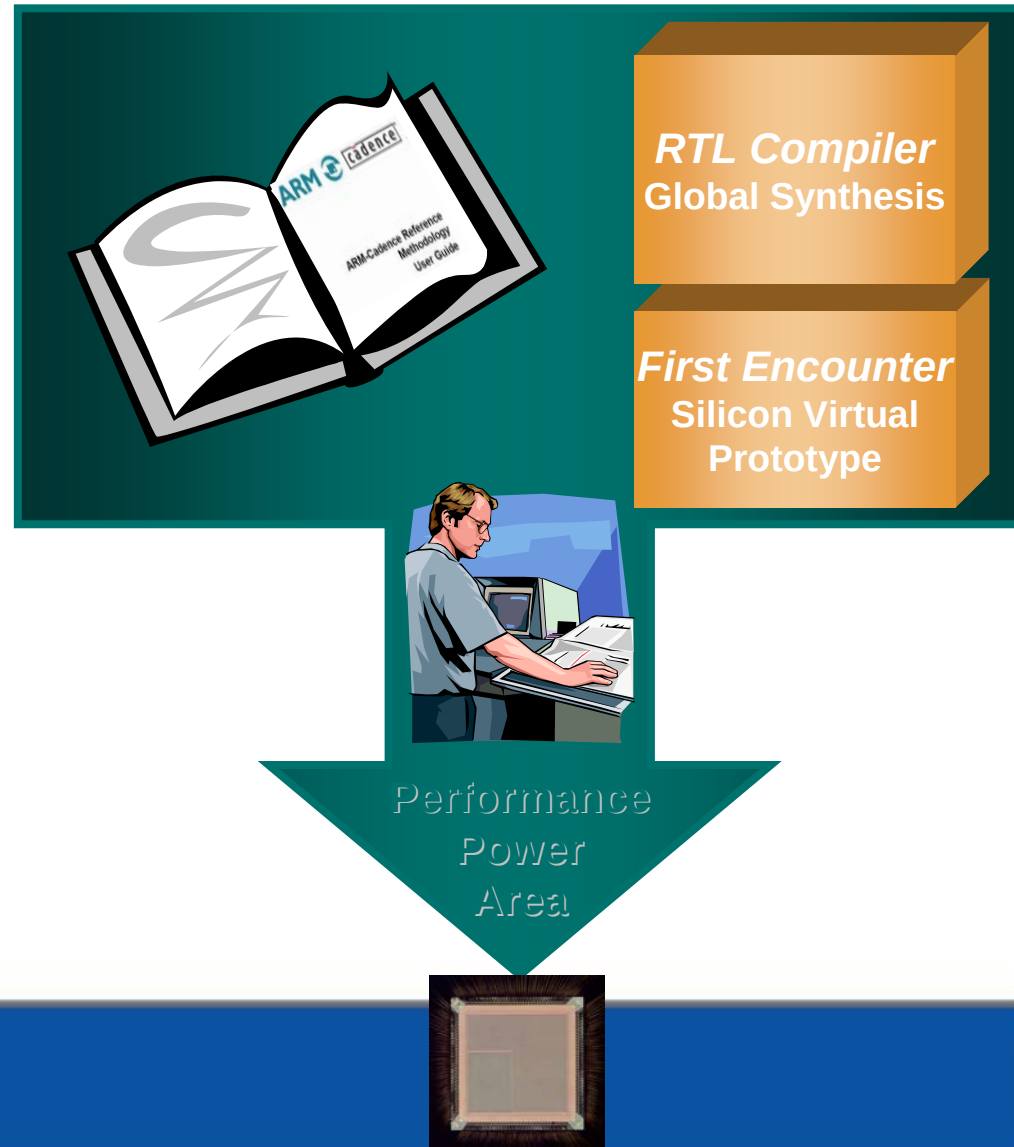
ARM-Cadence Reference Methodology for ARM1176-IEM

- Flow features
 - Automatic MSV (Multi Supply Voltage) implementation flow from RTL to GDS
 - Top-down, single-pass MSV synthesis, with concurrent leakage and dynamic power optimization
 - Automatic level shifter and clamp cell insertion and handling throughout the flow
 - Full Chip multi-domain signal integrity analysis
 - IR-drop aware sign-off static timing analysis
 - Functional and equivalence checking of multi domain implementation
- ARM reference libraries
 - Artisan Metro 130G + Power-Management Kit
- Availability
 - Base ARM1176JZ(F)-S support in Q106
 - IEM version support in H106



Cadence Optimization Methodology Kit for Improved PPA

- Methodology
 - Builds on ARM-Cadence Reference Methodology
 - Includes ARM Artisan Metro™ and Advantage front-end library views that support TSMC 90G and 130G processes
- Tools
 - Encounter RTL Compiler
 - First Encounter Virtual Prototyping
- Support
 - Customizes Reference Methodology for specific PPA goals
 - Includes training and follow-on support



ARM-Cadence Collaborate in Digital Design

- Focused programs
 - Low-power test chip
 - ARM-CDN Encounter Reference Methodology
 - Cadence Optimization Methodology Kit for ARM processors
- Delivering customer benefits
 - Improved time to market
 - Lower design risk
 - Flexible, differentiated design solutions

